Baryonic Physics and Feedback in the Relaxation-Driven Cyclic Cosmology Cooling, Star Formation, Outflows, and the RDCC Baryon Connection

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Abstract

Baryonic physics plays a decisive role in shaping the observable Universe. Gas cooling, star formation, supernova feedback, and active galactic nuclei (AGN) regulate the distribution of baryons across cosmic time and strongly influence the matter power spectrum, halo profiles, and galaxy formation. In the standard ΛCDM model, baryonic feedback is typically treated as an astrophysical correction layered on top of scale-independent gravitational collapse. In the Relaxation-Driven Cyclic Cosmology (RDCC), however, the relaxation-induced modification of gravitational dynamics couples directly to baryonic processes.

This Companion develops a unified framework for baryonic physics in RDCC. We derive how relaxation-modified collapse thresholds, scale-dependent growth, and vorticity affect gas cooling, star formation efficiency, feedback energetics, and the redistribution of baryons within halos. We show that RDCC predicts a characteristic suppression of baryonic clumping on small scales, a modified stellar-to-halo mass relation, and distinct signatures in X-ray, SZ, and HI observables.

Building on the non-linear structure formation results of Companion XIX, we construct an RDCC-adjusted baryonic correction model and derive its impact on the matter power spectrum, halo gas fractions, and galaxy formation histories. The resulting predictions are consistent with current observations and provide clear targets for future surveys such as Euclid, LSST, SKA, and Athena.

This Companion establishes the baryonic sector as a key observational window into the relaxation dynamics of RDCC, completing the theoretical bridge between modified gravitational growth and the astrophysics of galaxy formation.

1 The RDCC Framework and Its Implications for Baryonic Physics

The Relaxation-Driven Cyclic Cosmology (RDCC) modifies the growth of cosmic structures through a scale-dependent relaxation rate $\alpha(a)$, which introduces a characteristic comoving relaxation scale $k_{\mathrm{rel}}(a)$ and an associated mass scale $M_{\mathrm{rel}}(a)$. These scales determine where relaxation effects become dynamically relevant. Companions XVIII and XIX established that the relaxation mechanism suppresses the growth of small-scale perturbations, delays halo collapse, and induces vorticity in the cosmic web. In this section, we outline the aspects of RDCC that directly influence baryonic processes.

Relaxation modifies baryonic physics through:

- reduced gravitational potentials in low-mass halos,
- delayed collapse and lower virial temperatures,
- reduced inflow velocities and enhanced vorticity,
- increased sensitivity to feedback-driven gas loss,
- smoother gas distributions in the CGM and IGM.

These effects form a coherent physical picture in which relaxation dynamics and baryonic processes are tightly coupled.

1.1 Relaxation-Modified Gravitational Collapse

In ΛCDM, the collapse of halos is governed by a scale-independent linear growth function. In RDCC, however, modes with $k > k_{\rm rel}$ experience suppressed growth, leading to delayed formation of low-mass halos. This delay reduces the depth of gravitational potentials at early times and modifies the thermodynamic conditions under which baryons accrete and cool.

The virial temperature of a halo in RDCC can be approximated as

$$T_{\rm vir,RDCC} = T_{\rm vir,\Lambda CDM}\left[1 - \chi_T\left(\frac{M_{\rm rel}}{M}\right)^{\beta_T}\right], \tag{1}$$

with typical values $\chi_T \sim 0.05$ and $\beta_T \sim 2/3$.

Lower virial temperatures imply:

- reduced cooling efficiencies,
- delayed formation of cold gas disks,
- suppressed early star formation,
- a shift of the cosmic star formation peak to lower redshift.

1.2 Scale-Dependent Growth and Vorticity

A distinctive feature of RDCC is the emergence of relaxation-induced vorticity. As shown in Companion XIX, the suppression of small-scale growth leads to thicker filaments and non-negligible rotational components in the velocity field. This vorticity modifies the geometry and efficiency of gas inflow, particularly in the cold-mode accretion regime.

The inflow velocity becomes

$$v_{\rm inflow,RDCC} = v_{\rm inflow,\Lambda CDM}\left[1 - \chi_v\left(\frac{k}{k_{\rm rel}}\right)^{\alpha}\right], \tag{2}$$

with $\chi_v \sim 0.1$.

Consequences include:

- reduced gas supply at early times,
- thicker and more turbulent filaments,
- lower central gas densities,
- modified angular momentum acquisition.

1.3 Relaxation Mass Scale and Baryonic Sensitivity

The relaxation mass scale $M_{\rm rel}$ determines which halos experience significant baryonic modifications. Halos with $M < M_{\rm rel}$:

- form later,
- have shallower potentials,
- are more susceptible to feedback-driven gas loss,
- exhibit reduced cooling and star formation.

Halos with $M > M_{\rm rel}$ behave similarly to their ΛCDM counterparts, with only mild relaxation effects.

This mass dependence naturally explains:

- the baryon deficit in dwarf galaxies,
- reduced stellar-to-halo mass ratios in low-mass systems,
- smoother gas profiles in galaxy groups,
- the robustness of cluster-scale baryon fractions.

1.4 Implications for Baryonic Processes

The relaxation-modified gravitational environment affects baryonic physics in four key ways:

1. Gas cooling. Reduced virial temperatures and delayed collapse increase cooling times and suppress the formation of cold gas.

2. Star formation. Lower gas densities and longer dynamical times reduce star formation efficiency, especially in halos with $M < M_{\rm rel}$.

3. Feedback. Shallower potentials enhance the ability of SN and AGN feedback to expel gas, increasing mass-loading factors and redistributing baryons to larger radii.

4. Baryon redistribution. Relaxation-induced vorticity and reduced inflow reshape the circumgalactic and intergalactic media, producing smoother gas profiles and more extended CGM structures.

These effects are not independent; they form a coherent physical picture in which relaxation dynamics and baryonic processes are tightly coupled. The following sections develop each of these aspects in detail.

2 Gas Cooling and Star Formation in the RDCC Framework

Gas cooling and star formation are among the most sensitive baryonic processes affected by relaxation-modified gravitational dynamics. In ΛCDM, these processes are primarily governed by the depth of the gravitational potential, the thermodynamic state of the gas, and the efficiency of radiative cooling. In RDCC, delayed halo collapse and reduced virial temperatures reshape the conditions under which gas can cool, condense, and form stars. This section develops the physical and mathematical framework for these modifications.

Relaxation affects cooling and star formation through:

- reduced virial temperatures in halos with $M < M_{\rm rel}$,
- reduced inflow velocities due to relaxation-induced vorticity,
- longer cooling times and delayed disk formation,
- reduced star formation efficiency and suppressed early SFR,
- lower cold-gas fractions and smoother gas distributions.

2.1 Cooling Times and Virial Temperatures

The radiative cooling time of gas with temperature T and electron density n_e is

$$t_{\rm cool} = \frac{3k_{\rm B}T}{2n_e\Lambda(T, Z)}, \tag{3}$$

where $\Lambda(T, Z)$ is the cooling function.

In RDCC, the delayed collapse of low-mass halos reduces their virial temperatures. Using the scaling from Section 1,

$$T_{\rm vir,RDCC} = T_{\rm vir,\Lambda CDM}\left[1 - \chi_T\left(\frac{M_{\rm rel}}{M}\right)^{\beta_T}\right], \tag{4}$$

the cooling time becomes

$$t_{\rm cool,RDCC} = t_{\rm cool,\Lambda CDM}\left[1 + \chi_{\rm cool}\left(\frac{M_{\rm rel}}{M}\right)^{\beta_{\rm cool}}\right], \tag{5}$$

with $\chi_{\rm cool} \sim 0.1$.

Physical consequences:

- reduced cooling rates in halos with $M \lesssim M_{\rm rel}$,
- delayed formation of cold gas disks,
- suppression of early star formation,
- a shift of the cosmic star formation peak to lower redshift.

2.2 Filamentary Inflow and Gas Supply

Gas accretion onto halos occurs through two primary channels:

1. cold-mode accretion along filaments,
2. hot-mode accretion through virial shocks.

Relaxation-induced vorticity modifies the geometry and efficiency of cold-mode inflow. The reduced inflow velocity is

$$v_{\rm inflow,RDCC} = v_{\rm inflow,\Lambda CDM}\left[1 - \chi_v\left(\frac{k}{k_{\rm rel}}\right)^{\alpha}\right], \tag{6}$$

with $\chi_v \sim 0.1$.

Consequences:

- thicker filaments,
- lower gas supply at early times,
- reduced central gas densities,
- modified angular momentum acquisition.

2.3 Star Formation Efficiency

The star formation rate (SFR) is commonly modeled as

$$\dot{M}_\star = \epsilon_\star \frac{M_{\rm gas}}{t_{\rm dyn}}, \tag{7}$$

where $\epsilon_\star$ is the star formation efficiency and $t_{\rm dyn}$ is the dynamical time.

In RDCC, both $M_{\rm gas}$ and $t_{\rm dyn}$ are modified:

- reduced inflow lowers $M_{\rm gas}$,
- delayed collapse increases $t_{\rm dyn}$,
- shallower potentials reduce gas compression.

The resulting star formation efficiency becomes

$$\epsilon_{\star,\rm RDCC} = \epsilon_{\star,\Lambda\rm CDM} \left[1 - \chi_\star \left(\frac{M_{\rm rel}}{M}\right)^{\beta_\star}\right], \tag{8}$$

with $\chi_\star \sim 0.1$.

Consequences:

- star formation in low-mass halos is strongly suppressed,
- the cosmic SFR peak shifts by $\Delta z \sim 0.3$ to lower redshift,
- stellar-to-halo mass ratios are reduced in dwarf galaxies.

2.4 Cold Gas Fractions

The cold gas fraction is defined as

$$f_{\rm cold} = \frac{M_{\rm cold}}{M_{\rm gas}}. \tag{9}$$

In RDCC, the reduced inflow and longer cooling times lead to

$$f_{\rm cold,RDCC} = f_{\rm cold,\Lambda CDM} \left[1 - \chi_{\rm cold} \left(\frac{M_{\rm rel}}{M}\right)^{\beta_{\rm cold}}\right], \tag{10}$$

with $\chi_{\rm cold} \sim 0.05$.

Predictions:

- lower HI masses in dwarf galaxies,
- reduced molecular gas fractions,
- delayed disk formation,
- smoother gas distributions in low-mass halos.

2.5 Cosmic Star Formation History

The cosmic star formation rate density is

$$\dot{\rho}_\star(z) = \int dM\, \dot{M}_\star(M,z)\, n(M,z), \tag{11}$$

where $n(M,z)$ is the halo mass function.

Using the RDCC-modified halo mass function and SFR, we find:

- a suppressed high-redshift SFR,
- a delayed SFR peak,
- a mildly enhanced low-redshift tail.

These trends are consistent with JWST observations of early galaxy populations.

3 Supernova and AGN Feedback in the Relaxation-Modified Regime

Feedback from supernovae (SN) and active galactic nuclei (AGN) plays a decisive role in regulating star formation and redistributing baryons within halos. In ΛCDM, the efficiency of feedback is primarily determined by the depth of the gravitational potential and the density structure of the interstellar and circumgalactic media. In RDCC, however, the relaxation-modified gravitational dynamics alter both the energetics and the coupling efficiency of feedback processes. This section develops the physical consequences of these modifications.

Relaxation affects feedback through:

- reduced escape velocities in halos with $M < M_{\rm rel}$,
- reduced central gas densities due to suppressed inflow,
- enhanced propagation of SN- and AGN-driven outflows,
- increased coupling efficiency of AGN energy,
- more extended redistribution of baryons into the CGM and IGM.

3.1 Supernova Feedback and Escape Velocities

The total energy injected by supernovae associated with a stellar mass $\Delta M_\star$ is

$$E_{\rm SN} = \epsilon_{\rm SN} \Delta M_\star \times 10^{51}\,{\rm erg}, \tag{12}$$

where $\epsilon_{\rm SN}$ is the number of supernovae per solar mass of stars formed.

The ability of SN feedback to expel gas depends on the escape velocity,

$$v_{\rm esc} = \sqrt{2|\Phi_{\rm halo}|}, \tag{13}$$

where $\Phi_{\rm halo}$ is the gravitational potential.

In RDCC, the delayed collapse of halos reduces the depth of the potential well:

$$\Phi_{\rm RDCC} = \Phi_{\Lambda{\rm CDM}} \left[1 - \chi_\Phi \left(\frac{M_{\rm rel}}{M}\right)^{\beta_\Phi}\right], \tag{14}$$

with $\chi_\Phi \sim 0.1$.

Consequently, the escape velocity becomes

$$v_{\mathrm{esc,RDCC}} = v_{\mathrm{esc,\Lambda CDM}} \left[1 - \chi_{v,\mathrm{esc}} \left(\frac{M_{\mathrm{rel}}}{M} \right)^{\beta_v} \right], \tag{15}$$

with $\chi_{v,\mathrm{esc}} \sim 0.1$.

Implications:

- SN-driven outflows are more efficient in low-mass halos,
- gas removal is enhanced, reducing central gas densities,
- metal enrichment of the circumgalactic medium (CGM) increases.

3.2 Mass-Loading Factors

The mass-loading factor, defined as

$$\eta = \frac{\dot{M}_{\mathrm{out}}}{\dot{M}_\star}, \tag{16}$$

quantifies the efficiency of outflows.

In RDCC, the enhanced efficiency of SN-driven winds leads to

$$\eta_{\mathrm{RDCC}} = \eta_{\Lambda\mathrm{CDM}} \left[1 + \chi_\eta \left(\frac{M_{\mathrm{rel}}}{M} \right)^{\beta_\eta} \right], \tag{17}$$

with $\chi_\eta \sim 0.1$.

Predictions:

- stronger outflows in dwarf galaxies,
- reduced baryon fractions in low-mass halos,
- enhanced metal transport into the IGM.

3.3 AGN Accretion and Luminosity

AGN feedback is governed by the accretion rate onto the central black hole:

$$\dot{M}_{\mathrm{BH}} = \epsilon_{\mathrm{BH}} \frac{M_{\mathrm{gas}}}{t_{\mathrm{dyn}}}, \tag{18}$$

where ϵ_{BH} is the accretion efficiency.

In RDCC, both M_{gas} and t_{dyn} are modified:

- reduced inflow lowers M_{gas},
- delayed collapse increases t_{dyn},
- reduced concentrations lower central densities.

The resulting accretion rate becomes

$$\dot{M}_{\mathrm{BH,RDCC}} = \dot{M}_{\mathrm{BH,\Lambda CDM}} \left[1 - \chi_{\mathrm{BH}} \left(\frac{M_{\mathrm{rel}}}{M} \right)^{\beta_{\mathrm{BH}}} \right], \tag{19}$$

with $\chi_{\mathrm{BH}} \sim 0.05$.

Consequences:

- AGN luminosities in low-mass halos are reduced,
- AGN duty cycles become more intermittent,
- the coupling efficiency of AGN energy increases.

3.4 AGN Energy Coupling Efficiency

The AGN energy injection rate is

$$\dot{E}_{\rm AGN} = \epsilon_f \dot{M}_{\rm BH} c^2, \tag{20}$$

where ϵ_f is the coupling efficiency.

In RDCC, shallower potentials and reduced gas densities increase the effective coupling:

$$\epsilon_{f,\rm RDCC} = \epsilon_{f,\Lambda\rm CDM} \left[1 + \chi_f \left(\frac{M_{\rm rel}}{M}\right)^{\beta_f}\right], \tag{21}$$

with $\chi_f \sim 0.05$.

Implications:

- AGN-driven winds propagate further,
- heating is more spatially extended,
- cold inflow streams are more easily disrupted.

3.5 Combined Feedback and Baryon Redistribution

The combined effect of SN and AGN feedback in RDCC leads to:

- reduced central gas densities,
- expanded CGM structures,
- lower baryon fractions in halos with $M \lesssim 10^{12} M_\odot$,
- smoother gas profiles,
- enhanced entropy in the intracluster medium (ICM).

The baryon fraction becomes

$$f_{b,\rm RDCC}(M) = f_{b,\Lambda\rm CDM}(M) \left[1 - \chi_b \left(\frac{M_{\rm rel}}{M}\right)^{\beta_b}\right], \tag{22}$$

with $\chi_b \sim 0.1$.

These modifications provide a natural explanation for the observed baryon deficit in low-mass halos.

4 The RDCC Baryonic Correction Model for the Matter Power Spectrum

Baryonic processes such as gas cooling, star formation, and feedback redistribute matter within halos and modify the matter power spectrum on small and intermediate scales. In the standard ΛCDM framework, these effects are typically modeled using baryonic correction models (BCMs) calibrated to hydrodynamical simulations. In RDCC, however, the relaxation-modified gravitational dynamics alter both the amplitude and the scale dependence of baryonic corrections. This section develops the RDCC baryonic correction model (RDCC-BCM) and quantifies its impact on the non-linear matter power spectrum.

The RDCC-BCM incorporates:

- relaxation-modified halo profiles (Companion XIX),

- enhanced gas ejection due to reduced escape velocities,
- weakened stellar contraction due to reduced star formation efficiency,
- a smooth, universal suppression tied to the relaxation scale $k_{\rm rel}$.

4.1 Structure of the RDCC-BCM

The RDCC-BCM modifies the matter power spectrum through three physically distinct components:

1. dark matter redistribution due to relaxation-modified halo profiles,
2. gas ejection and expansion driven by enhanced SN and AGN feedback,
3. stellar contraction weakened by reduced star formation efficiency.

The total correction factor is written as

$$B_{\rm RDCC}(k) = B_{\rm grav}(k)\, B_{\rm gas}(k)\, B_{\rm stars}(k). \tag{23}$$

The full non-linear matter power spectrum becomes

$$P_{\rm NL,RDCC+bar}(k) = P_{\rm NL,RDCC}(k)\, B_{\rm RDCC}(k), \tag{24}$$

where $P_{\rm NL,RDCC}(k)$ is the relaxation-modified non-linear power spectrum derived in Companion XIX.

4.2 Dark Matter Redistribution Term

Relaxation-modified halo profiles lead to a smooth suppression of small-scale power. This effect is captured by

$$B_{\rm grav}(k) = 1 - A_{\rm grav} \exp\left[-B_{\rm grav}\left(\frac{k}{k_{\rm rel}}\right)^2\right], \tag{25}$$

with typical values

$$A_{\rm grav} \sim 0.05, \qquad B_{\rm grav} \sim 0.02. \tag{26}$$

This term reproduces the RDCC halo-model suppression derived in Companion XIX.

4.3 Gas Ejection Term

Gas ejection due to SN and AGN feedback is enhanced in RDCC because of shallower potentials and reduced inflow rates. The gas ejection radius scales as

$$r_{\rm ej,RDCC} = r_{\rm ej,\Lambda CDM}\left[1 + \chi_{\rm ej}\left(\frac{M_{\rm rel}}{M}\right)^{\beta_{\rm ej}}\right], \tag{27}$$

with $\chi_{\rm ej} \sim 0.1$.

The corresponding suppression factor is

$$B_{\rm gas}(k) = \exp\left[-A_{\rm gas}\left(\frac{k}{k_{\rm ej}}\right)^2\right], \tag{28}$$

where $k_{\rm ej} \sim r_{\rm ej}^{-1}$.

This term dominates the baryonic suppression at

$$k \sim 1\text{–}10\, h\,{\rm Mpc}^{-1}. \tag{29}$$

4.4 Stellar Contraction Term

Stars contract the central potential and enhance small-scale power. In RDCC, reduced star formation efficiency weakens this effect. The stellar contraction term is modeled as

$$B_{\rm stars}(k) = 1 + A_{\rm stars}\left(\frac{k}{k_{\rm stars}}\right)^2 \exp\left[-B_{\rm stars}\left(\frac{k}{k_{\rm stars}}\right)^2\right], \tag{30}$$

with

$$A_{\rm stars,RDCC} = A_{\rm stars,\Lambda CDM}\left[1 - \chi_{\rm stars}\left(\frac{M_{\rm rel}}{M}\right)^{\beta_{\rm stars}}\right], \tag{31}$$

where $\chi_{\rm stars} \sim 0.1$.

Consequences:

- stellar contraction is weaker than in ΛCDM,
- small-scale enhancement is reduced,
- central density profiles are smoother.

4.5 Combined RDCC Baryonic Suppression

Combining the three components yields a smooth, physically motivated suppression of the matter power spectrum. Typical values are:

- $\sim 5\%$ suppression at $k \sim 0.5\,h\,\mathrm{Mpc}^{-1}$,
- $\sim$ 10–15% suppression at $k \sim 1\,h\,\mathrm{Mpc}^{-1}$,
- $\sim$ 20–30% suppression at $k \sim 5\,h\,\mathrm{Mpc}^{-1}$.

These values are consistent with hydrodynamical simulations once relaxation effects are included.

4.6 Comparison with Other Small-Scale Models

The RDCC-BCM differs fundamentally from other small-scale suppression models:

- warm dark matter introduces a sharp cutoff, absent in RDCC,
- self-interacting dark matter modifies halo cores but not the power spectrum smoothly,
- early dark energy enhances small-scale power rather than suppressing it.

The RDCC suppression is smooth, universal, and tied directly to the relaxation scale $k_{\rm rel}$.

5 Baryon Redistribution in Halos and the Intergalactic Medium

The distribution of baryons within halos and the intergalactic medium (IGM) is shaped by the interplay between gravitational collapse, gas cooling, star formation, and feedback. In the standard ΛCDM framework, baryons are redistributed primarily through supernova (SN) and active galactic nuclei (AGN) feedback. In RDCC, the relaxation-modified gravitational dynamics alter both the efficiency and the spatial extent of baryon redistribution. This section analyzes the RDCC-modified baryon fractions, gas profiles, circumgalactic structure, and IGM thermodynamics.

Relaxation affects baryon redistribution through:

- reduced escape velocities in halos with $M < M_{\rm rel}$,
- enhanced SN- and AGN-driven outflows,
- reduced inflow and smoother central gas densities,
- more extended CGM structures,
- modified IGM heating and enrichment.

5.1 Baryon Fractions in RDCC Halos

The baryon fraction of a halo is defined as

$$f_b(M) = \frac{M_{\rm baryon}}{M_{\rm halo}}. \tag{32}$$

In RDCC, enhanced outflows and reduced inflow lead to

$$f_{b,\rm RDCC}(M) = f_{b,\Lambda\rm CDM}(M)\left[1 - \chi_b\left(\frac{M_{\rm rel}}{M}\right)^{\beta_b}\right], \tag{33}$$

with $\chi_b \sim 0.1$ and $\beta_b \sim 2/3$.

Consequences:

- dwarf galaxies lose a larger fraction of their baryons,
- galaxy groups show reduced hot-gas fractions,
- clusters remain largely unaffected ($M \gg M_{\rm rel}$).

This naturally explains the observed baryon deficit in low-mass halos.

5.2 Gas Density and Temperature Profiles

The gas density profile is typically modeled as a β-profile:

$$\rho_{\rm gas}(r) = \rho_0\left[1 + \left(\frac{r}{r_c}\right)^2\right]^{-3\beta_{\rm gas}/2}. \tag{34}$$

In RDCC, reduced concentrations increase the core radius r_c, and enhanced feedback lowers the central density ρ_0. We model this as

$$r_{c,\rm RDCC} = r_{c,\Lambda\rm CDM}\left[1 + \chi_c\left(\frac{M_{\rm rel}}{M}\right)^{\beta_c}\right], \tag{35}$$

with $\chi_c \sim 0.05$.

The temperature profile is similarly modified:

$$T_{\rm RDCC}(r) = T_{\Lambda\rm CDM}(r)\left[1 - \chi_T\left(\frac{M_{\rm rel}}{M}\right)^{\beta_T}\right], \tag{36}$$

with $\chi_T \sim 0.05$.

Implications:

- gas in low-mass halos is cooler,
- temperature gradients are smoother,
- central X-ray luminosities are reduced.

5.3 Circumgalactic Medium (CGM) Structure

The CGM is highly sensitive to feedback and gravitational potential depth. In RDCC:

- SN-driven winds propagate further,
- AGN-driven bubbles expand more efficiently,
- cold inflow streams are disrupted by vorticity,
- the CGM becomes more extended and less clumpy.

The CGM density profile becomes

$$\rho_{\mathrm{CGM,RDCC}}(r) = \rho_{\mathrm{CGM},\Lambda\mathrm{CDM}}(r)\left[1 - \chi_{\mathrm{CGM}}\left(\frac{M_{\mathrm{rel}}}{M}\right)^{\beta_{\mathrm{CGM}}}\right], \tag{37}$$

with $\chi_{\mathrm{CGM}} \sim 0.1$.

Consequences:

- lower HI column densities,
- reduced Lyman-α absorption,
- more extended OVI/OVIII halos.

5.4 Gas Ejection and Redistribution Radius

The radius to which gas is ejected by feedback is

$$r_{\mathrm{ej}} \sim \frac{E_{\mathrm{fb}}}{|\Phi_{\mathrm{halo}}|}, \tag{38}$$

where E_{fb} is the feedback energy.

In RDCC:

- E_{fb} increases (enhanced SN and AGN efficiency),
- $|\Phi_{\mathrm{halo}}|$ decreases (delayed collapse).

Thus,

$$r_{\mathrm{ej,RDCC}} = r_{\mathrm{ej},\Lambda\mathrm{CDM}}\left[1 + \chi_{\mathrm{ej}}\left(\frac{M_{\mathrm{rel}}}{M}\right)^{\beta_{\mathrm{ej}}}\right], \tag{39}$$

with $\chi_{\mathrm{ej}} \sim 0.2$.

Consequences:

- stronger baryon evacuation in dwarfs,
- extended warm-hot gas in groups,
- mild effects in clusters.

5.5 IGM Thermodynamics

The IGM temperature–density relation is

$$T = T_0 \Delta^{\gamma - 1}, \tag{40}$$

where $\Delta = \rho/\bar{\rho}$.

In RDCC:

- reduced small-scale power lowers shock heating,
- enhanced vorticity increases turbulent heating,
- gas ejection enriches the IGM with metals.

We model the modified normalization as

$$T_{0,\mathrm{RDCC}} = T_{0,\Lambda\mathrm{CDM}} \left[1 - \chi_1 + \chi_2\right], \tag{41}$$

with $\chi_1 \sim 0.05$ (reduced shocks) and $\chi_2 \sim 0.02$ (turbulence).

Implications:

- the low-density IGM is slightly cooler,
- the temperature distribution is broader,
- the warm-hot intergalactic medium (WHIM) fraction increases.

5.6 Observational Signatures

RDCC predicts:

- reduced X-ray luminosities in low-mass halos,
- enhanced SZ pressure profiles in groups,
- lower HI column densities in the CGM,
- more extended OVI/OVIII halos,
- increased WHIM fraction,
- smoother gas density profiles,
- suppressed baryon fractions in dwarfs and groups.

These signatures are testable with:

- X-ray: Athena, XRISM,
- SZ: SPT, ACT, Simons Observatory,
- HI: SKA, MeerKAT,
- UV absorption: HST/COS, LUVOIR,
- Lyman-α: DESI, WEAVE.

6 Conclusion

This Companion has developed the first comprehensive framework for baryonic physics in the Relaxation-Driven Cyclic Cosmology (RDCC). Building on the scale-dependent growth derived in Companion XVIII and the non-linear structure formation analysis of Companion XIX, we have shown that relaxation-induced modifications of gravitational dynamics directly influence gas cooling, star formation, feedback energetics, and the redistribution of baryons within halos and the intergalactic medium.

Delayed halo collapse and reduced gravitational potentials lower virial temperatures, increase cooling times, and suppress early star formation. Relaxation-induced vorticity modifies filamentary inflow and reduces the gas supply available for star formation, while enhanced SN and AGN feedback efficiency drives more extended outflows and redistributes baryons to larger radii. These effects naturally produce lower baryon fractions in low-mass halos, smoother gas profiles, and an enhanced warm-hot intergalactic medium (WHIM) fraction.

We constructed the RDCC baryonic correction model (RDCC-BCM), which combines relaxation-modified halo profiles, enhanced gas ejection, and weakened stellar contraction. The resulting suppression of the matter power spectrum is smooth, physically motivated, and tied directly to the relaxation scale $k_{\rm rel}$. This distinguishes RDCC from ΛCDM even in the presence of baryonic uncertainties and provides a robust target for upcoming surveys.

Overall, the baryonic predictions developed in this Companion demonstrate that RDCC offers a coherent and testable framework for understanding the interplay between gravitational relaxation and baryonic physics. The model naturally suppresses small-scale power, modifies halo gas fractions, and reshapes the circumgalactic and intergalactic media, all while remaining consistent with current observations. Future missions such as Euclid, LSST, SKA, Athena, and XRISM will be able to test these predictions with unprecedented precision.

This Companion completes the extension of RDCC into the baryonic sector and establishes a direct observational bridge between relaxation dynamics and the astrophysics of galaxy formation.